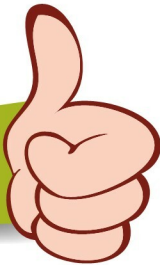


The Perfect Prefect



In this article, we shall discuss NAGIOS, an open source software that is deployed in most data centres to monitor various system and network parameters.

The practice of appointing prefects is a time-honoured one. Whether it is in schools, colleges, armies or societies, from time immemorial, overseers (or prefects) like district magistrates or religious priests, have played an important part in regulating and monitoring the performance and day-to-day activities of the groups of people they oversee.

Information technology is no different—an overseer/monitor is required to constantly regulate and monitor the health of the hardware, software and network in modern-day data centres. These ‘prefects’ of the data centre provide vital information to systems administrators—such as the amount of free disk space, network outages, application and hardware downtime, and even server-room temperatures.

NAGIOS (a recursive acronym for “Nagios Ain’t Gonna Insist On Sainthood”) has been one of the most favoured ‘prefects’ of the data centre, monitoring parameters such as systems status (whether a system is up and running; CPU/memory/disk usage, etc), service status (whether a service is up and running—DNS, Web server, mail server, etc), and many other factors including room temperature and even humidity! It can generate alerts (through email/SMS) when the monitored parameters exceed pre-set thresholds.

As I sit down to write this article, I am glad to share with you that this ‘perfect prefect’ has saved many of my clients hundreds of hours of downtime. Just recently, a customer decided to move a problematic database from a central database server host, because Nagios had alerted us about a possible problem with one of the schemas, which was adversely affecting the overall health of the database server, and could have severely affected other mission-critical production database schemas on the same host.

In this article, I shall try to dispel a commonly held myth—that Nagios is difficult to install. I distinctly remember that about five years back, a senior manager in a big IT firm called me and mentioned this as one of the reasons why the

company planned to outsource the installation to us. It might have required a bit of tweaking then, but now that is no more the case—you can easily install and configure it to meet your requirements. Let us install and configure Nagios to monitor a sample service, and hence get an idea of how Nagios can benefit you and your organisation.

Let us install Nagios on an RHEL 5 host called prefect.knaff.org. We will use it to monitor itself—whenever it is available—and send alerts to nagios-admin@localhost in case of an outage. In the next article of this series, we will look at monitoring remote hosts and services.

Installation

On Red Hat Enterprise Linux, Nagios can be easily installed using the EPEL Repository. To the uninitiated, EPEL is: “Extra Packages for Enterprise Linux (or EPEL), a Fedora Special Interest Group that creates, maintains, and manages a high-quality set of additional packages for Enterprise Linux, including, but not limited to, Red Hat Enterprise Linux (RHEL), CentOS and Scientific Linux (SL).” More details can be obtained from <http://fedoraproject.org/wiki/EPEL>.

To ensure that Nagios is available in the EPEL repository, let’s browse the relevant repository (since ours is a 64-bit host, the x86_64 EPEL repository is at http://download.fedoraproject.org/pub/epel/5/x86_64/repoview/). On jumping to packages whose names begin with ‘N’, we can see that (as of this writing), there are 65 Nagios packages (RPMs) available for 64-bit RHEL 5. We can check this using the following command (on the URL for the group of packages we just mentioned):

```
[vbg@vbg ~]$ elinks --dump http://download.fedoraproject.org/pub/epel/5/x86_64/repoview/letter_n.group.html | grep -i nagios |
grep -v html | wc -l
65
[vbg@vbg ~]$
```